Abstract Algebra III: Assignment - Week 10

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June 11, 2026

Problem 1

Let R be a ring and V be an R-module. Let I be an ideal of R. Suppose that d_I is defined on V and on R. Then

(i) Addition is continuous from $V \times V$ to V.

(ii) Multiplication is continuous from $R \times V$ to V.

(iii) If d_I is defined on the R-module W and $f \in \mathrm{Hom}_R(V, W)$, then f is continuous.

Solution: (1) Let the addition map be $f : V \times V \to V$, where $f(x, y) = x + y$. Let O be an arbitrary open set in the target space V. We need to prove that its preimage $f^{-1}(O)$ is an open set in $V \times V$. If $f^{-1}(O)$ is empty, it is trivially open. If $f^{-1}(O)$ is non-empty, pick an arbitrary point $(x, y) \in f^{-1}(O)$. It means $x + y \in O$. Since O is an open set in V, there exists an integer n such that:

$$(x + y) + I^n V \subseteq O.$$

Let $U_x = x + I^n V$ and $U_y = y + I^n V$ be the open neighborhood of x, y respectively. Then $U_x \times U_y$ is an open neighborhood of (x, y) in $V \times V$. Since

$$f(U_x \times U_y) = U_x + U_y = (x + I^n V) + (y + I^n V) = x + y + I^n V$$

we have $f(U_x \times U_y) \subseteq O$ which implies

$$U_x \times U_y \subseteq f^{-1}(O).$$

Therefore, for arbitrary $(x, y) \in f^{-1}(O)$, we have an open neighborhood $U_x \times U_y$ contained in $f^{-1}(O)$. It turns out that $f^{-1}(O)$ is open and thus f is continuous.

(2) Let the multiplication map be $g : R \times V \to V$, where $g(r, v) = rv$. Let O be an arbitrary open set in the target space V. We need to prove that its preimage $g^{-1}(O)$ is an open set in $R \times V$. If $g^{-1}(O)$ is empty, it is trivially open. If $g^{-1}(O)$ is non-empty, pick an arbitrary point $(r, v) \in g^{-1}(O)$. It means $rv \in O$. Since O is an open set in V, there exists an integer n such that:

$$rv + I^n V \subseteq O.$$

Let $U_r = r + I^n$ and $U_v = v + I^nV$ be the open neighborhood of r, v respectively. Then $U_r \times U_v$ is an open neighborhood of (r, v) in $R \times V$. Since

$$g(U_r \times U_v) = U_rU_v = (r + I^n)(v + I^nV) \subseteq rv + r(I^nV) + (I^n)v + I^{2n}V \subseteq rv + I^nV$$

we have $g(U_r \times U_v) \subseteq O$ which implies

$$U_r \times U_v \subseteq g^{-1}(O).$$

Therefore, for arbitrary $(r, v) \in g^{-1}(O)$, we have an open neighborhood $U_r \times U_v$ contained in $g^{-1}(O)$. It turns out that $g^{-1}(O)$ is open and thus g is continuous.

(3) Let O be an arbitrary open set in the target space W. We need to prove that its preimage $f^{-1}(O)$ is an open set in V. If $f^{-1}(O)$ is empty, it is trivially open. If $f^{-1}(O)$ is non-empty, pick an arbitrary point $v \in f^{-1}(O)$. It means $f(v) \in O$. Since O is an open set in W, there exists an integer n such that:

$$f(v) + I^nW \subseteq O.$$

Let $U_v = v + I^nV$ be the open neighborhood of v in V. Since f is an R-module homomorphism, we have $f(I^nV) = I^nf(V) \subseteq I^nW$. Then

$$f(U_v) = f(v + I^nV) = f(v) + f(I^nV) \subseteq f(v) + I^nW$$

we have $f(U_v) \subseteq O$ which implies

$$U_v \subseteq f^{-1}(O).$$

Therefore, for arbitrary $v \in f^{-1}(O)$, we have an open neighborhood U_v contained in $f^{-1}(O)$. It turns out that $f^{-1}(O)$ is open and thus f is continuous. ■

Problem 2

The following algebra isomorphisms hold for R-algebras A and B, where R is a commutative ring.

(i) $M_n(A) \otimes_R B \simeq M_n(A \otimes_R B)$.

(ii) $M_n(A) \otimes_R M_m(B) \simeq M_{nm}(A \otimes_R B)$.

Solution: (i) Let

$$\begin{aligned} \phi : M_n(A) \times B &\to M_n(A \otimes_R B) \\ ((a_{ij}), b) &\mapsto (a_{ij} \otimes b) \end{aligned}$$

Since ϕ is R-bilinear, it induces a well-defined R-linear map

$$\begin{aligned} \Phi : M_n(A) \otimes_R B &\to M_n(A \otimes_R B) \\ (a_{ij}) \otimes b &\mapsto (a_{ij} \otimes b) \end{aligned}$$

For any $(a_{ij}), (c_{ij}) \in M_n(A)$ and $b, d \in B$, we have

$$\Phi(((a_{ij}) \otimes b)((c_{ij}) \otimes d)) = \Phi((a_{ij})(c_{ij}) \otimes bd) = \left(\sum_k a_{ik}c_{kj} \otimes bd\right).$$

On the other hand,

$$\Phi((a_{ij})\otimes b)\Phi((c_{ij})\otimes d) = (a_{ij}\otimes b)(c_{ij}\otimes d) = \left(\sum_k (a_{ik} \otimes b)(c_{kj} \otimes d)\right) = \left(\sum_k a_{ik}c_{kj} \otimes bd\right).$$

Thus Φ preserves multiplication and is an algebra homomorphism. The map

$$\Psi : M_n(A \otimes_R B) \to M_n(A) \otimes_R B$$
$$(a \otimes b)E_{ij} \mapsto aE_{ij} \otimes b$$

provides a two-sided inverse to Φ. Therefore, we obtain that

$$M_n(A) \otimes_R B \simeq M_n(A \otimes_R B).$$

(ii) By applying the isomorphism from (i) and treating $M_m(B)$ as an R-algebra, we have

$$M_n(A) \otimes_R M_m(B) \simeq M_n(A \otimes_R M_m(B)).$$

Since R is commutative, the tensor product is symmetric up to isomorphism, yielding

$$A \otimes_R M_m(B) \simeq M_m(B) \otimes_R A.$$

Applying (i) again to $M_m(B) \otimes_R A$, we get

$$M_m(B \otimes_R A) \simeq M_m(A \otimes_R B).$$

Substituting this sequence of isomorphisms back gives:

$$M_n(A \otimes_R M_m(B)) \simeq M_n(M_m(A \otimes_R B)).$$

For any ring S, there is a natural ring isomorphism $M_n(M_m(S)) \simeq M_{nm}(S)$ given by viewing an $n \times n$ block matrix of $m \times m$ matrices as an $nm \times nm$ matrix. Letting $S = A \otimes_R B$, we conclude

$$M_n(A) \otimes_R M_m(B) \simeq M_{nm}(A \otimes_R B).$$

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